CSA1516- Cloud Computing and Big data Analytics

CO1 - AT3 (Short Answer Test)

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Slot: A

1) Define cloud computing and explain any four characteristics that make it different from traditional onpremise computing.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

3) Explain the role of server virtualization in cloud computing Differentiate between virtualization as an enabling technology and cloud computing as a service model.

4) Discuss the role of data centers in delivering cloud services.

5) Differentiate Iaas, Paas, Saas and Xaas with one suitable real world example for each model.

Answers

1) Definition:

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking and software over the internet on a pay as you use basis.

Four characteristics:

1. On demand self service: users can access computing resources whenever without human interaction with service provider.

2. Broad Network access: cloud services can be accessed through the internet using laptops, mobiles and tablets

3. Resource Pooling: computing resources are shared among multiple users using a multi tenant model.

4. Rapid elasticity: Resources can be increased or decreased quickly according to demand.

5. Difference:

Traditional on premise computing requires organizations to purchase and maintain their own hardware, where as cloud computing provides flexible resources over internet with lower initial costs.

2) Cloud computing development through several stages

Mainframe Computers:

large computers shared by multiple users through terminals.

Personal Computers:

Computing became available in individual users

Distributed Computing:

Multiple computers worked together to share tasks and resources.

Virtualization:

one physical server could run multiple virtual

Machines, improving hardwares utilization.

Internet & web development:

High speed internet & web applications allowed users to access software remoteler.

Modern cloud Platform:

Provides such as Aws, Microsoft Azure & Google cloud began offering servers, storage databases & software over the internet.

* Hardware improvement, virtualization & internet based software enabled modern cloud Platform to provide scalable, flexible & cost effective services.

3) Role of Server Virtualization:

Server virtualization divides one physical servers into multiple virtual machine. Each virtual machine can ron its own operating system and applications. It improves resource utilization reduces hardware costs & allows flexible resource allocation.

Virtualization:

* It is a technology that creates virtual versions
* It allows multiple virtual machines to run on physical layer.
* It can operate within a private organization.

Ex: VMware, ESXi

Cloud computing:

* It delivers computing resources as services over a network.
* It provides on demand access to servers, storage etc.
* It can be public, private or hybrid.

Ex: Amazon web services (AWS).

4) Definition:

A data center is a facility containing servers, storage systems, networking equipment, and supporting infrastructure used to process and store data.

Roles of data centers:

1. Data storage: storage files, databases, and application data
2. processing: Provides computing power to run applications and servers.
3. Networking: connects servers to the internet and enables users to access and services.
4. Security: uses access controls, firewalls, and monitoring to protect data.
5. Availability: uses backup systems, redundant power, and cooling to maintain server availability.
6. Scalability: Allows cloud providers to add computing and storage resources as demand increases.

Service model	Meaning	What it provides
Iaas	Infrastructure as a service	Virtual machines, Storage and networking
Paas	Platform as a service	A platform to develop, test and deploy applications.
Saas	Service software as a service	Ready to use software over internet
Xaas	Anything as a service.	Various IT resources and services delivered over a network

Iaas : users manage the operating system and applications, while the provider manages the underlying infrastructure.

Paas : Developers build and deploy applications without managing the underlying servers.

Saas ; user access complete software applications through a browser or APP

Xaas : A broad team for delivering different services over the internet, such as storage, security and database.